

# 實習生月報

## PDO 資料路徑重建

CoE / SDO 交易重建

PDO Mapping / Assignment 關聯整理

SyncManager / FMMU 資料路徑分析

報告期間

2026.08

MONTHLY REPORT

# 本月工作大綱



7/30 - 8/5

## 理解封包與 EtherCAT 100%

- 使用 Wireshark 觀察從上電抓的原始 EtherCAT 封包
- 整理 Broadcast、AP、FP 等指令對 Slave 的操作方式
- 整理 Master 對 Slave ESC 暫存器的讀寫流程
- 理解 CoE、SDO、PDO 與 Object Dictionary 的關係
- 建立本月主線：CoE / SDO → PDO Mapping / Assignment → SyncManager → FMMU

8/6 - 8/12

## 建立 Slave 拓樸關係 80%

- 理解 Master 如何透過 Auto Increment / Configured Address 指定 Slave
- 理解 Configured Station Address 與 Configured Station Alias 的差異
- 設計 AP Datagram 配對、Slave Discovery 模型與分析流程

## 設計 Analyzer 驗證 Agent 20%

- 建立 Stage 3 Result Check
- 由 Capture evidence 驗證 Slave Count、Topology、Address 與 Identity

8/13 - 8/19

## Mailbox / CoE / SDO 70%

- 在 Datagram Record 中加入 CoE / SDO decoded fields
- 重建 SDO Initiate Download Request / Response Transaction
- 依 SDO Index / SubIndex 辨識 PDO Mapping / Assignment 操作
- 重建 RxPDO / TxPDO Mapping Configuration
- 重建 0x1C12 / 0x1C13 PDO Assignment Configuration

## 完善 Agent 驗證流程 30%

- 整理 TShark EtherCAT / CoE / SDO 查詢與 JSON 結構規格
- 將常用 Capture 查詢封裝成 Python tools，提供 LLM 多種驗證方式

8/20 - 8/21

## SyncManager / FMMU 分析 100%

- 在 Datagram Record 中加入 SyncManager / FMMU decoded fields
- 重建 SyncManager Configuration
- 重建 FMMU Configuration
- 驗證 PDO Assignment 與 SM2 / SM3 的設定關係
- 驗證 FMMU Physical Address 與 SyncManager RAM 區段的對應

SECTION / 01

# 01

## 上電流程與 ESC Register 解析

追蹤啟動封包，建立 ESC 暫存器存取與解析脈絡

# 上電封包操作流程

## 01 Broadcast Initialization

BRD / BWR / BRW  
All Slaves

## 02 Auto Increment Addressing

APRD / APWR / APRW  
Topology Position

## 03 Configured Station Address

APWR / ADO 0x0010  
Fixed Address

## 04 Fixed Position ESC Access

FPRD / FPWR / FPRW  
Configured Address + ADO

### Master

#### Slave 1

Outgoing ADP: 0x0000  
Signed: 0  
Topology Position: 1

#### Slave 2

Outgoing ADP: 0xFFFF  
Signed: -1  
Topology Position: 2

#### Slave 3

Outgoing ADP: 0xFFFE  
Signed: -2  
Topology Position: 3

| Outgoing ADP | Signed Value | 命中的拓樸位置 |
|--------------|--------------|---------|
| 0x0000       | 0            | Slave 1 |
| 0xFFFF       | -1           | Slave 2 |
| 0xFFFE       | -2           | Slave 3 |

Master 以 outgoing ADP = 0、-1、-2... 逐站指定 Slave。  
ADO 是目標 ESC Register Offset，命中的 Slave 會執行該 Datagram。

# SII EEPROM 讀取流程

## 01 檢查 EEPROM 狀態

- 讀取 ESC Register **0x0502~0x0503**
- 確認 Busy 清除且沒有 Error

## 02 指定 SII Word Address

- 寫入 ESC Register **0x0504~0x0507**
- **0x0008** = Vendor ID · **0x000A** = Product Code 起始 Word

## 03 送出 EEPROM Read Command

- 在 **0x0502~0x0503** 的 Control / Status 欄位送出 Read Command

## 04 讀取 EEPROM Data

- 以 APRD 讀取 **ADO = 0x0508**
- 依前面指定的 SII Word Address 解讀資料內容

| Address Space   | Address              | Meaning                 |
|-----------------|----------------------|-------------------------|
| ESC Register    | <b>0x0502~0x0503</b> | EEPROM Control / Status |
| ESC Register    | <b>0x0504~0x0507</b> | EEPROM Address          |
| ESC Register    | <b>0x0508~0x050F</b> | EEPROM Data             |
| SII EEPROM Word | <b>0x0008~0x0009</b> | Vendor ID               |
| SII EEPROM Word | <b>0x000A~0x000B</b> | Product Code            |

## 實際 Capture

實際 Capture 中，Master 會以一個 **APWR** Datagram，從 **ADO = 0x0502** 開始一次寫入 **6 Bytes**。前 2 Bytes 對應 **0x0502~0x0503** 的 EEPROM Control / Status，後 4 Bytes 對應 **0x0504~0x0507** 的 EEPROM Address，因此一次 **APWR** 即可同時送出 Read Command 與欲讀取的 SII Word Address。

EEPROM Read 完成後，Master 再以 **APRD** 讀取 **ADO = 0x0508** 的 EEPROM Data。實際取得的資料內容，則由前面寫入的 SII Word Address 決定，例如 **0x0008** 對應 Vendor ID、**0x000A** 對應 Product Code。

# Slave Discovery 重建成果

## 01 重點統整

AP Commands

## 02 AP-Analyzer 重建流程

Slave Discovery

## 03 程式碼實作

Class Relationship

## 04 實際成果

Stage 3 Result

### Slave Discovery

| AP Datagram

| Topology Position

| Configured Address

| Slave Identity

SECTION / 02

# 02

## CoE / SDO 與 PDO Mapping 重建

從 CoE / SDO Transaction 重建 PDO Mapping 與 Assignment，建立 Process Data 的資料定義。



# CoE / SDO 知識整理

01 Mailbox / CoE / SDO  
Protocol Layers

02 Object Dictionary  
Index / SubIndex

03 SDO Download / Upload  
Client / Server

04 SDO → PDO  
Configuration  
PDO Mapping

## CoE / SDO 知識主線

EtherCAT Datagram

Mailbox

CoE

SDO

Object Dictionary

PDO Configuration

CoE / SDO 的作用是透過 EtherCAT Mailbox 存取 CANopen Object Dictionary，並可進一步用來設定 PDO Mapping / Assignment。



# SDO Transaction / PDO Mapping 重建成果

EtherCATAnalyzer.TestConsole.exe analyze

Stage 4 / 105 Transactions

[Stage 4] SDO Transaction Analysis

Elapsed: 6.226 s

Transaction Count: 105

Configured Slave Address: 0x0001, Object: 0x1C12:00, Data: 0x00, Request Frame: 41394, Response Frame: 41403, Success: True, Abort Code: N/A  
Configured Slave Address: 0x0001, Object: 0x1C13:00, Data: 0x00, Request Frame: 41408, Response Frame: 41417, Success: True, Abort Code: N/A  
Configured Slave Address: 0x0001, Object: 0x1A00:00, Data: 0x00, Request Frame: 41422, Response Frame: 41455, Success: True, Abort Code: N/A  
Configured Slave Address: 0x0001, Object: 0x1A00:01, Data: 0x60410010, Request Frame: 41460, Response Frame: 41614, Success: True, Abort Code: N/A  
Configured Slave Address: 0x0001, Object: 0x1A00:02, Data: 0x60430010, Request Frame: 41639, Response Frame: 41786, Success: True, Abort Code: N/A  
Configured Slave Address: 0x0001, Object: 0x1A00:03, Data: 0x30221110, Request Frame: 41791, Response Frame: 41958, Success: True, Abort Code: N/A  
Configured Slave Address: 0x0001, Object: 0x1A00:04, Data: 0x30210c10, Request Frame: 41963, Response Frame: 42130, Success: True, Abort Code: N/A  
Configured Slave Address: 0x0001, Object: 0x1A00:05, Data: 0x30210210, Request Frame: 42135, Response Frame: 42302, Success: True, Abort Code: N/A  
Configured Slave Address: 0x0001, Object: 0x1A00:00, Data: 0x05, Request Frame: 42307, Response Frame: 42316, Success: True, Abort Code: N/A  
Configured Slave Address: 0x0001, Object: 0x1A01:00, Data: 0x00, Request Frame: 42321, Response Frame: 42330, Success: True, Abort Code: N/A  
Configured Slave Address: 0x0001, Object: 0x1A02:00, Data: 0x00, Request Frame: 42335, Response Frame: 42344, Success: True, Abort Code: N/A  
Configured Slave Address: 0x0001, Object: 0x1A03:00, Data: 0x00, Request Frame: 42349, Response Frame: 42376, Success: True, Abort Code: N/A  
Configured Slave Address: 0x0001, Object: 0x1600:00, Data: 0x00, Request Frame: 42381, Response Frame: 42496, Success: True, Abort Code: N/A  
Configured Slave Address: 0x0001, Object: 0x1600:01, Data: 0x60420010, Request Frame: 42501, Response Frame: 42676, Success: True, Abort Code: N/A  
Configured Slave Address: 0x0001, Object: 0x1600:00, Data: 0x01, Request Frame: 42681, Response Frame: 42690, Success: True, Abort Code: N/A  
Configured Slave Address: 0x0001, Object: 0x1601:00, Data: 0x00, Request Frame: 42695, Response Frame: 42704, Success: True, Abort Code: N/A  
Configured Slave Address: 0x0001, Object: 0x1601:01, Data: 0x60400010, Request Frame: 42709, Response Frame: 42842, Success: True, Abort Code: N/A  
Configured Slave Address: 0x0001, Object: 0x1601:02, Data: 0x60600008, Request Frame: 42847, Response Frame: 43014, Success: True, Abort Code: N/A  
Configured Slave Address: 0x0001, Object: 0x1601:03, Data: 0x60720010, Request Frame: 43019, Response Frame: 43193, Success: True, Abort Code: N/A



# SDO / PDO Analyzer 實作設計

## 01 設計流程

Reconstruction Pipeline

## 02 程式碼關係

Class Relationship

SDO Transaction

PDO Mapping

PDO Assignment

SECTION / 03

# 03

## SyncManager 與 Local RAM 資料交換

解析 SyncManager Configuration，確認 PDO Process Data 實際使用的 ESC Local RAM Buffer。

# SyncManager 功能與 PDO Assignment 關係

01

## SyncManager

Local RAM Control

02

## RxPDO Path

0x1C12 → SM2

03

## TxPDO Path

0x1C13 → SM3

PDO Mapping



PDO Assignment



SyncManager



ESC Local RAM

0x1C12 / 0x1C13 決定目前使用哪些 PDO Mapping ;  
SyncManager 則管理 Process Data 在 ESC Local RAM 中的實際資料區域與存取方式。



# SyncManager Analyzer 實作設計

## 01 設計流程

Reconstruction Pipeline

## 02 程式碼關係

Class Relationship

### SyncManager Configuration Reconstruction

SM Register → SyncManager Configuration → Local RAM Layout

# SyncManager Analyzer 重建成果

## [PDO Assignment Summary]

Station Address: 0x0001  
Assignment Object: 0x1C12  
Type: RxPDO Assignment  
Active Entries: 1

| Entry | Mapping Object |
|-------|----------------|
|-------|----------------|

|       |       |
|-------|-------|
| ----- | ----- |
|-------|-------|

|    |        |
|----|--------|
| 01 | 0x1600 |
|----|--------|

Station Address: 0x0001  
Assignment Object: 0x1C13  
Type: TxPDO Assignment  
Active Entries: 1

| Entry | Mapping Object |
|-------|----------------|
|-------|----------------|

|       |       |
|-------|-------|
| ----- | ----- |
|-------|-------|

|    |        |
|----|--------|
| 01 | 0x1A00 |
|----|--------|

## [SyncManager Summary]

Station Address: 0x0001  
SyncManager: SM2

SECTION / 04

# 04

## FMMU 與 Logical Address Mapping

解析 FMMU Configuration，建立 EtherCAT Logical Address 與 ESC Physical RAM 的映射關係。



# FMMU Analyzer 實作設計

01

## FMMU

Logical Mapping

FMMU ( Fieldbus Memory Management Unit ) 是 ESC 內的硬體單元，負責把 EtherCAT Logical Address 空間映射到 ESC 的 Physical / Local RAM 區域。Master 因此能以連續邏輯位址讀寫 Process Data，不必直接處理各 Slave 內部 RAM 的實體位址。

PDO Assignment 決定使用哪些 PDO Mapping；SyncManager 管理這批 Process Data 實際使用的 Local RAM Buffer；FMMU 再把該 RAM 區域映射到 EtherCAT Logical Address。

PDO Mapping / Assignment → SyncManager → FMMU → Process Data

02

## 設計流程

Reconstruction Pipeline

03

## 程式碼關係

Class Relationship

04

## 重建成果

Analyzer Result

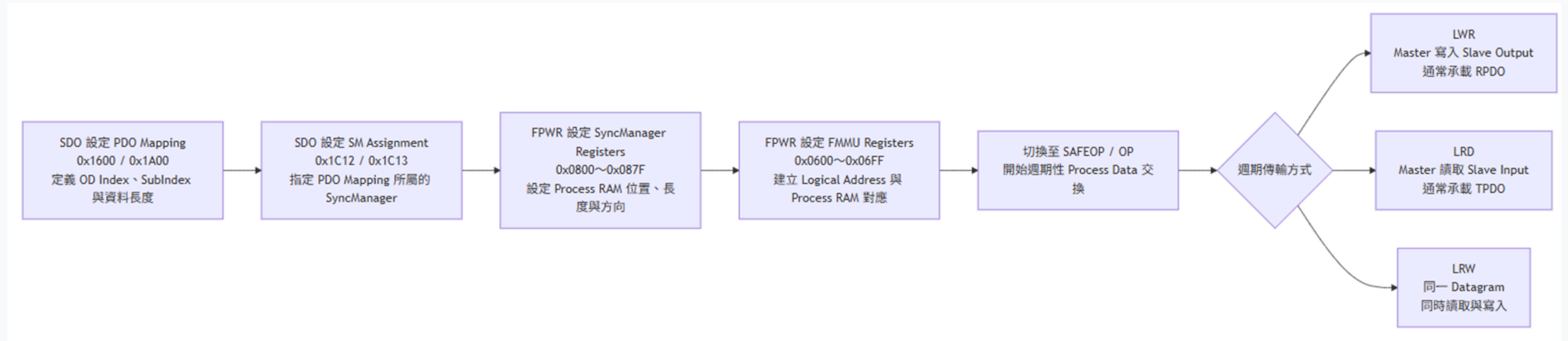
| Field                   | Meaning                                    |
|-------------------------|--------------------------------------------|
| Logical Start Address   | EtherCAT logical address space 的映射起點       |
| Length                  | 此 FMMU mapping 涵蓋的資料長度                     |
| Physical Start Address  | 對應 ESC Local RAM 的起始位址                     |
| Direction / Orientation | 定義 logical read / write 與 local access 的方向 |
| Enabled / Activate      | 啟用或停用此 FMMU mapping                        |

SyncManager 管理實體 Local RAM，FMMU 管理 Logical Address 到 Local RAM 的映射。兩者串接後，Master 才能透過 LRW、LRD、LWR 等 Process Data 命令操作 PDO 資料。





# 總結



# EtherCAT 圖解分析

FIGURE / 04



## [ Diagram Canvas ]

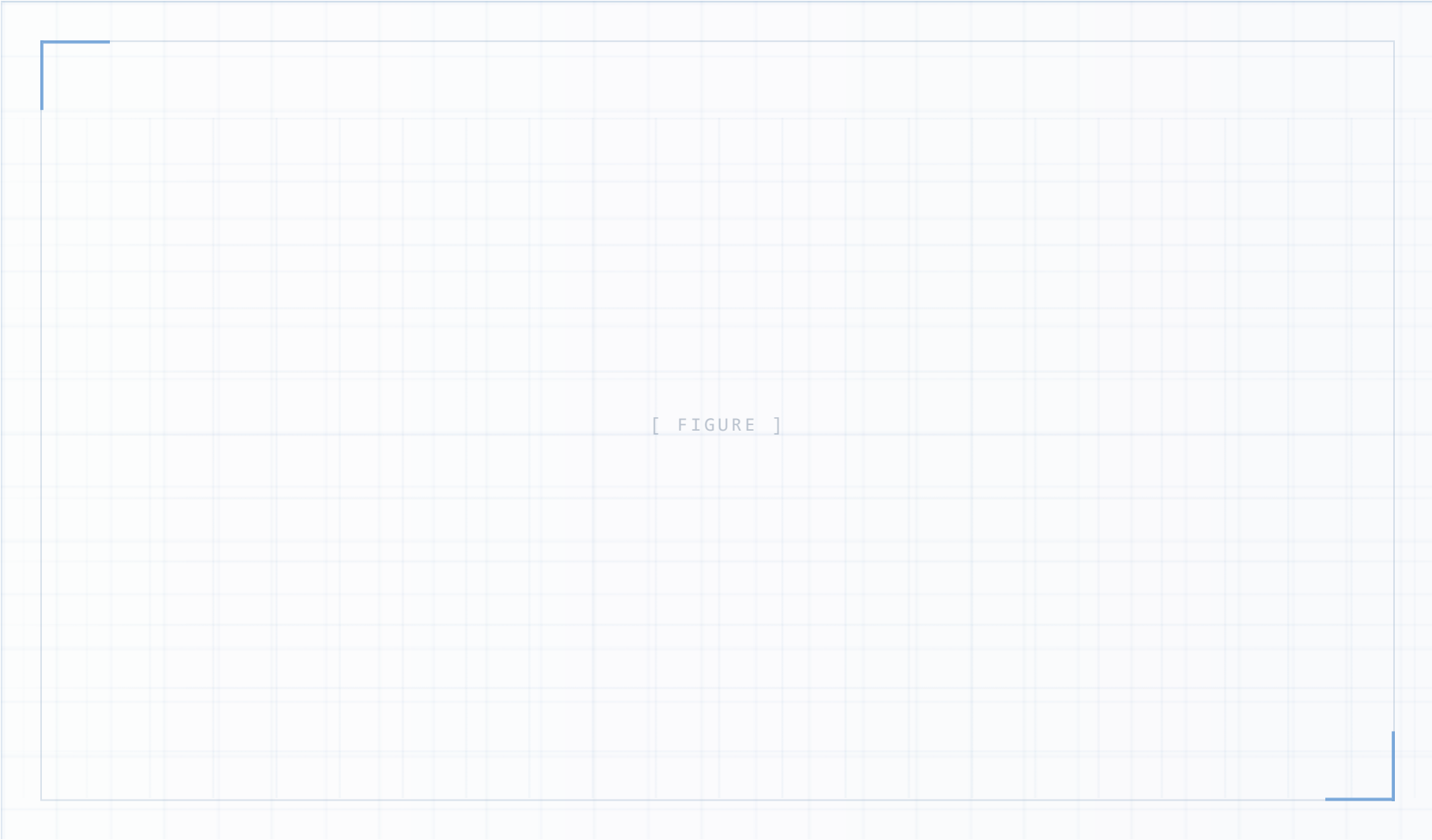
Class Diagram / Architecture Diagram / Data Path Figure

REPLACE WITH IMG / SVG / FIGURE / HTML DIAGRAM

### FOCUS POINTS

- 01 Diagram Purpose**  
Explain the main relationship between modules.
- 02 Key Components**  
Highlight the major classes or architecture blocks.
- 03 Reading Order**  
Describe how to read the diagram efficiently.
- 04 Implementation Relevance**  
Connect the diagram to the current development work.

# EtherCAT 資料路徑概覽



## KNOWLEDGE POINTS

01

### FMMU

Logical Address  
→ ESC local

02

### SyncManager

Consistent data  
exchange control

03

### Process RAM

Stores mapped  
process data

04

### DATA PATH

Logical Address → FMMU →  
SM → RAM → Application

